

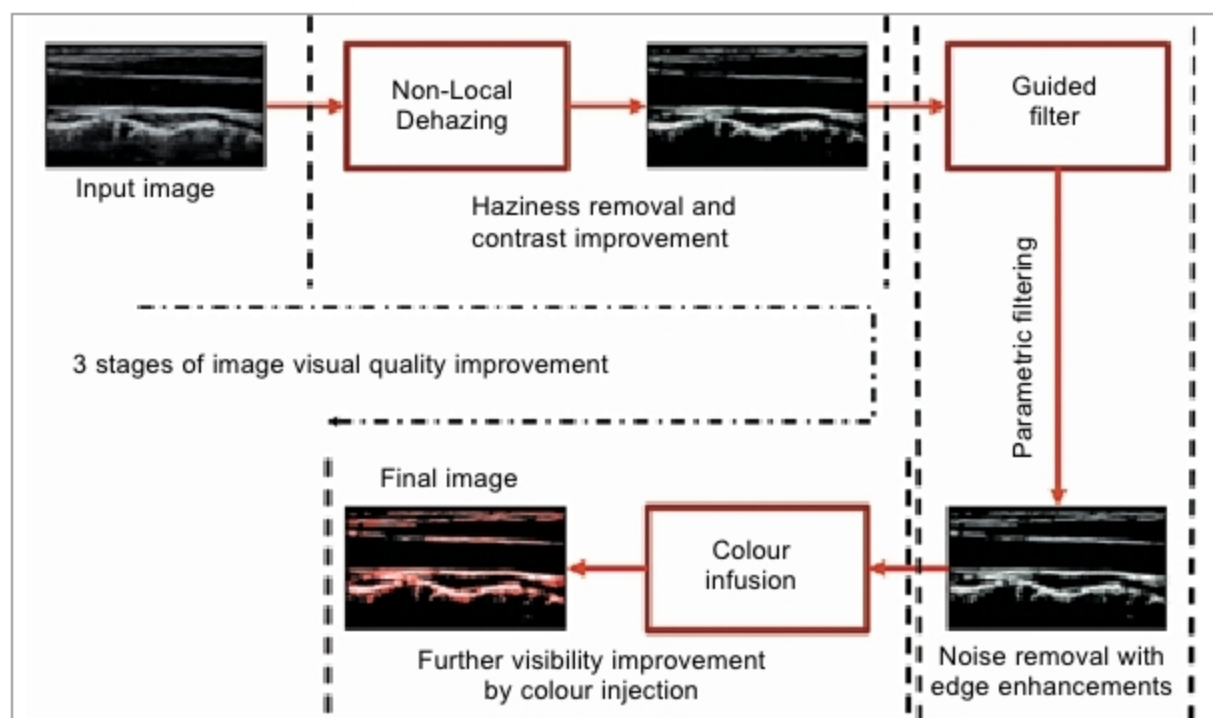


Fig. 2: Ultrasound image super-resolution

decreases with the distance as the ultrasonic wave diffracts. It depends directly upon the near-zone length. Near-zone length is inversely proportional to wavelength of the acoustic wave. It also depends upon the frequency, width and damping factor of the transducer.

Applications

Ultrasounds can be used in many applications in the field of medical diagnosis, such as:

- Fetus size measurement to determine the due date of delivery
- To check number of fetuses, the position of placenta and the sex of the baby
- For checking inside the heart for any cardiovascular ailment
- To measure blood flow (Doppler mode) through heart and kidney
- To detect prostate cancer in kidney
- To detect stones in gall bladder or kidney
- To detect tumors in ovaries and breasts
- To detect the fatty liver by observing its size

It is also used together with other modalities to do some complex medical analysis of the patient.

Pros and cons of ultrasound imaging

Ultrasound imaging possesses some advantages over other imaging modalities like:

- First and foremost being that, it is a non-invasive method that does not

require any cut in the patient's body.

- It does not expose patient's body to any harmful radiations like x-rays or CT scans, which is very important for pregnant mothers.
- It captures soft tissues very well, unlike x-rays.
- It is becoming less expensive and more accessible with rapid miniaturisation.

But ultrasound imaging has some disadvantages also. For instance, it has been found that pregnant mothers having frequent ultrasound examinations may develop risk of delivering underweight babies. And that the local temperature of tissues increases due to frequent collision with ultrasonic energy.

Besides, there are some challenges

that still exist in this field. One such challenge, addressed in this article, is the low spatial resolution of ultrasound images. In addition, ultrasonic images are also prone to Gaussian and speckle noise. There are some artifacts, like mirror image artifacts, which also impact ultrasound imaging.

To address all these challenges, similar strategies can be adopted such that maximum number of problems can be solved at minimal cost.

Some recent trends

Ultrasonic imaging has evolved a lot in the past few decades. Performance enhancement and addition of new features have frequently been done in this field. Rapid commercialisation of hand-held ultrasonic imaging devices has made this modality even more popular. The smaller devices are easy to carry and so can be taken to remote rural places for point-of-care diagnosis.

Work has been going on to reduce the noise in the ultrasonic images. For that, Tissue Harmonic imaging has been introduced in which reconstruction of image is done using second harmonic of echo from the tissue.

Recently, 3D and 4D ultrasonic imaging has also been introduced to further enhance the experience of visualisation of motion of the heart wall. The 4D technology is aimed at competing with imaging modalities like CT and MRI in terms of resolution and image clarity. Incorporating ultrasonic equipment with GPS has

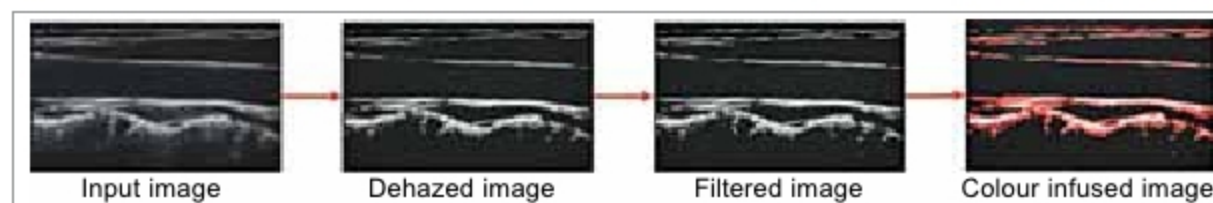


Fig. 3: Results for dataset 1

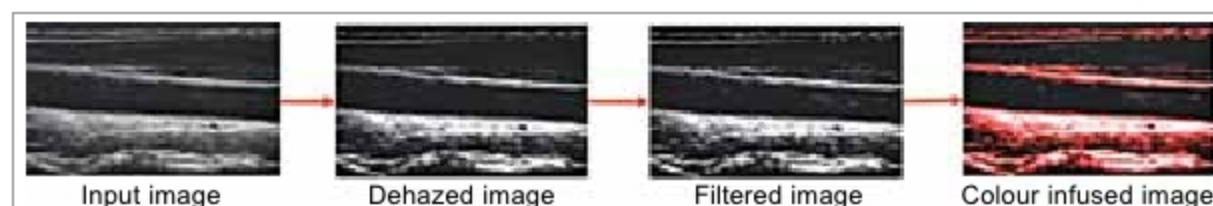


Fig. 4: Results for dataset 2

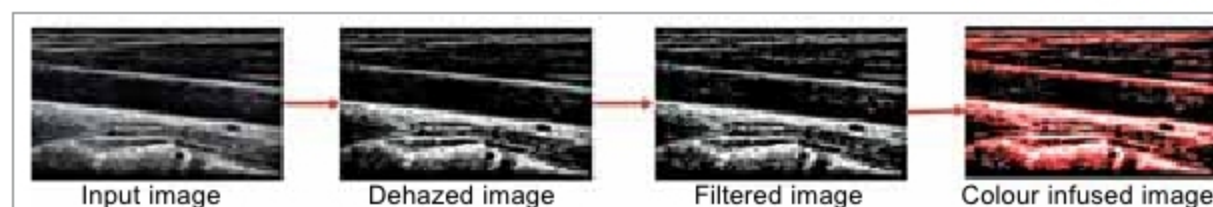


Fig. 5: Results for dataset 3